

Aarav Lee

Embedded Systems & IoT Firmware Engineer • 11 Years Experience • MERIQ Index: 87% (Strong Candidate)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: FreeRTOS, Python, PCB Debugging, C/C++, MQTT, I2C/SPI/UART

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Qualcomm* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core FreeRTOS service, reducing latency by 25%.
- Led a squad of 4 engineers delivering scalable microservices processing 15M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 55%.

Associate Embedded Systems & IoT Firmware Engineer — *Intel* (2015 - 2023)

- Designed and maintained Python data endpoints with 98% SLA uptime.
- Optimized database indexing and caching layers using PCB Debugging saving \$16k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [FreeRTOS, Python, PCB Debugging, C/C++]
Engineered end-to-end production solution leveraging FreeRTOS and Python with full test automation.

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [PCB Debugging, C/C++, MQTT, I2C/SPI/UART]
Architected high-resilience system integrating PCB Debugging with 99.9% uptime.

EDUCATION & CREDENTIALS

M.S. in Electrical and Computer Engineering

Vellore Institute of Technology (VIT) — 2011 - 2015 (Score: 3.74 / 4.0)